pmsimstats-ng Codebase Overview

pmsimstats team

2026-04-09

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The `pmsimstats-ng` R package implements Monte Carlo simulation-based power analysis for aggregated N-of-1 clinical trial designs, comparing four trial designs on their ability to detect a biomarker-treatment interaction. The package descends from the publication code of Hendrickson et al. (2020), *Frontiers in Digital Health* 2:13. Since the merge on 2026-04-08, the package supports two data-generating-process architectures: Architecture B (multivariate normal differential correlation, the original Hendrickson approach) and Architecture A (direct mean moderation), selectable via the `dgp_architecture` parameter. The dual-architecture distinction is the subject of the white paper `02-dgp-mean-moderation-vs-mvn.tex`.

1 Core R Package Files (`R/`)

1.1 generateData.R

The data-generating process (DGP). Three exported functions:

- **`generateData()`** – Draws simulated participant data from a multivariate normal distribution. Accepts a pre-built covariance matrix (`cached_sigma`) or builds one via `buildSigma()`. Uses a pre-computed Cholesky factor for efficient MVN draws (`L %*% Z + mu`). Computes outcome columns via vectorized `data.table` operations. The `dgp_architecture` parameter selects between Architecture B (`"mvn"`, default) and Architecture A (`"mean_moderation"`); under Architecture A, the BM-BR correlation block is omitted from Σ and the biomarker-treatment interaction is applied as an additive post-draw step $\mathrm{BR}_{it} \mathrel{+}= c_{bm} \cdot B_i^* \cdot \mathbf{1}[\text{on drug}] \cdot \sigma_{\mathrm{BR}}$, where B_i^* is the standardized biomarker.
- **`buildSigma()`** – Constructs the 26x26 covariance matrix for a single trial path. Also accepts `dgp_architecture`: under Architecture A the BM-BR correlation block is set to zero. Implements:
  - AR(1) within-factor autocorrelation: $\rho^{|t_i - t_j|}$
  - Cross-factor correlations (same-time and different-time with AR(1) decay)
  - Biomarker-BR correlation with exponential decay during off-drug periods: $c_{bm} \cdot \exp(-\lambda_{cor} \cdot t_{sd})$ (Architecture B only)
  - Auto-derives $\lambda_{cor} = \ln(2)/$`carryover_t1half`
  - Positive definiteness check with verbose warning
  - Pre-computes Cholesky factor for caching
- **`validateParameterGrid()`** – Tests all parameter combinations for positive definiteness before simulation begins, reporting failures upfront. Note: this function currently builds sigma matrices with Architecture B's BM-BR correlation block regardless of the simulation's actual `dgp_architecture` setting, so it may report spurious PD failures when running Architecture A with high `c.bm`. The data generation itself is unaffected; the warning can be safely ignored under Architecture A. See doc `19` for the full discussion.

1.2 generateSimulatedResults.R

The simulation loop. Single exported function:

- **`generateSimulatedResults()`** – Orchestrates the full parameter sweep:
  1. PD pre-validation of all parameter combinations
  2. Sigma matrix caching (builds all unique matrices once)
  3. Parameter grid expansion across designs, response params, baseline params, and model params
  4. Per-parameter-set worker function (`run_one_paramset`) that generates data, fits LME, applies censoring, and collects results

5. Parallel execution via `furrr::future_map()` with closure-based function passing to workers
6. Progressive save support for long runs

Key parameters:

- `dgp_architecture`: `"mvn"` (default) or `"mean_moderation"`.
- `lambda_cor`: correlation decay rate; default `NA` (auto-derived from `carryover_t1half`).
- `n_cores`: parallelism; default auto-detect.

1.3 lme_analysis.R

The analysis model. Single exported function:

- **`lme_analysis()`** – Fits a linear mixed-effects model to simulated or actual trial data:
  - Uses `nlme::lme` with `corCAR1(form = ~t|ptID)` for continuous-time AR(1) residual correlation
  - Dynamic formula selection: `Sx ~ bm + t + Dbc + bm:Dbc` when within-subject drug variation exists; `Sx ~ bm + t + bm:t` for open-label
  - `Dbc` is a continuous drug indicator (1 when on drug, exponential decay when off drug)
  - Graceful handling of rank-deficient models (returns NA)
  - Fallback to `lme` without `corCAR1` if optimization fails
  - Extracts interaction coefficient from `nlme` tTable

1.4 carryover_analysis.R

Extended analysis functions for actual trial data:

- **`characterize_carryover()`** – Sweeps candidate carryover half-lives (default 0.1 to 8 weeks), fits `nlme::lme` at each, compares AIC/BIC to identify the best-fitting half-life. Returns drug effect, interaction, AR(1) phi at each half-life.
- **`analyze_trial_extended()`** – Extracts both the main drug effect (`Dbc`) and biomarker interaction (`bm:Dbc`) with 95% CIs. Computes variance components, predicted drug effect as a function of biomarker value, and clinical decision threshold.
- **`print_carryover_summary()`** / **`print_trial_summary()`** – Clinical reporting summaries.

1.5 buildtrialdesign.R

Trial design specification:

- **`buildtrialdesign()`** – Constructs trial paths from intuitive inputs (timepoints, expectancies, on-drug vectors). Computes derived timing variables: `tod` (time on drug), `tsd` (time since discontinuation), `tpb` (time in positive belief).

1.6 utilities.R

Helper functions:

- **`modgompertz()`** – Modified Gompertz function for response trajectories. Passes through origin, asymptotes at `maxr`.
- **`cumulative()`** – Converts interval durations to cumulative time.
- **`reknitsimresults()`** – Recombines progressive save chunks.

1.7 censordata.R

Dropout simulation:

- **`censordata()`** – Two-component dropout: time-dependent (flat rate) and outcome-dependent (biased by symptom change). Implements last-observation-carried-forward censoring.

1.8 plottingfunctions.R

Visualization:

- **`PlotModelingResults()`** – Power heatmaps across 4-dimensional parameter grids with `ggplot2` `geom_tile` and `facet_grid`.
- **`plotfactortrajectories()`** – Factor trajectory plots by trial path.

2 Implementation Collections (`implementations/`)

In addition to the active package code in `R/`, the repository contains three parallel implementations of the simulation framework, each supporting different DGP architectures and coding styles. The active `R/` directory mirrors `implementations/original-extended/`.

Collection	Architectures	Style
`original`	B (MVN) only	data.table
`original-extended`	A + B	data.table
`tidyverse`	A + B	tidyverse

- **`original`** – Corrected Hendrickson publication code, restricted to Architecture B. Useful as a backward-compatibility baseline and for testing the Hendrickson et al. (2020) specification in isolation. Has no `dgp_architecture` parameter.
- **`original-extended`** – Production implementation. Adds the `dgp_architecture` parameter to all relevant functions, supporting both Architecture A (`"mean_moderation"`) and Architecture B (`"mvn"`, default). The active `R/` package directory matches this collection file-for-file.
- **`tidyverse`** – Complete tidyverse reimplementation. All routines live in a single `functions.R` file and use snake_case names (e.g., `generate_data`, `build_sigma_matrix`, `lme_analysis`). Supports both architectures via the same `dgp_architecture` parameter.

Each collection has its own `README.md` documenting parameter usage, validation results, and any collection-specific behavior. All three accept the same core parameters (`c.bm`, `carryover_t1half`, `lambda_cor`); only `original-extended` and `tidyverse` accept the `dgp_architecture` parameter.

3 Analysis Scripts

3.1 Figure 4 (`analysis/figure4/`)

Script	Purpose
`01-generate-power-sweep.R`	Figure 4: current code, sequential
`02-generate-power-sweep-original.R`	Figure 4: original commit code
`03-generate-power-by-commit.R`	Figure 4: parameterized by commit
`04-generate-power-head.R`	Figure 4: current code, parallel, c.bm={0,0.25,0.45}
`05-pd-diagnostics.R`	PD failure rate instrumentation
`06-plot-power-heatmaps.R`	Heatmap plotting with provenance

3.2 Figure 5 (`analysis/figure5/`)

Script	Purpose
`01-generate-parameter-sensitivity.R`	Figure 5: revised code
`02-generate-parameter-sensitivity-original.R`	Figure 5: original code

3.3 Extracted Code by Commit

Code from each historical commit extracted for comparison. Following the 2026-04-08 reorganization, these snapshots live in `analysis/archive/`:

```
analysis/archive/commit_42ac030/   # Initial (publication)
analysis/archive/commit_8609f12/   # Ron Thomas edits
analysis/archive/commit_f6ee86d/   # Autocorrelation typo fix
analysis/archive/commit_f70f86d/   # Carryover added
analysis/archive/commit_325314f/   # Modified Db (broken)
analysis/archive/original_R/       # Copy of 42ac030
```

The earlier exploratory tidyverse pipeline (`analysis/2025/`) has been moved to an external archive at `~/Dropbox/prj/alz/10-pmsimstats-ng/archive/2025/` and is no longer part of the working repository. Its successor is the maintained tidyverse implementation in `implementations/tidyverse/` (see Section 2).

4 Documentation (`docs/`)

The full documentation index is in `00-documentation-index.tex`; the table below summarizes each document by number. Sources are in `docs/`; rendered PDFs are in `docs/pdf/`.

#	Document	Content
01	Codebase Overview	Package structure, parameters, dependencies (this doc)
02	Two DGP Architectures	White paper: Architecture A vs. B power under carryover
03	Audit and Revision Report	Seven DGP corrections, PD resolution, deliverables
04	Revised Power Analysis	Architecture B problems, fixes, validated results
06	AR(1) Residual Correlation	`lmer` to `nlme` transition with Type I validation
07	Positive Definiteness Failures	PD failure prevalence and causes (Arch B)
08	BM-BR Correlation Decay	Pharmacokinetic derivation of λ_{cor} (Arch B)
09	Carryover Correlation Artifact	BM-BR correlation artifact analysis (Arch B)
10	Carryover Analysis Assessment	Carryover characterization in actual data
13	Figure 4 Walkthrough	Pre-audit methodology reference (historical)
14	Trial Analysis Guide	Collaborator goal implementation
15	Tidyverse Alignment Plan	Planning record (largely executed in April 8 merge)
16	Tidyverse Code Review	Review of pre-merge `analysis/2025/` code (historical)
17	Simplification Plan	Exploratory; not adopted in April 8 refactor
18	Response Parameter Sensitivity	Figure 5 sensitivity before/after (Arch B)
19	Mean Moderation Implementation	Architecture A derivation (`.tex` canonical, `.md` summary)
20	Annotated Bibliography	N-of-1 simulation methodology literature
21	Architecture Comparison Summary	Brief empirical contrast of Arch A vs. B
22	Journal Target Recommendation	Publication strategy for white paper `02`

File	Content
`results_core.rda`	Published Figure 4 results (initial commit, 500 reps)
`extracted_bp.rda`	Baseline parameters (BP mean/SD, CAPS mean/SD)
`extracted_rp.rda`	Response parameters (Gompertz max/disp/rate/sd)
`CTdata.rda`	Actual clinical trial data
`results_maxes.rda`	Figure 5A results (response maxima)
`results_rates.rda`	Figure 5B results (response rates)
`results_trajectories.rda`	Factor trajectory data

5 Bundled Data (`data/`)

6 Key Parameters

6.1 Current Configuration (Revised Code)

6.2 Publication Configuration (Original Code)

7 Dependencies

Core: `data.table`, `nlme`, `lme4`, `lmerTest`, `corpcor`, `MASS`, `ggplot2`

Performance: `furrr`, `future` (parallel processing)

Legacy (in DESCRIPTION but used minimally): `svMisc`, `tictoc`, `ggpubr`, `gridExtra`

Parameter	Value	Rationale
`dgp_architecture`	`"mvn"` (default)	Architecture B; `"mean_moderation"` for A
Autocorrelation	AR(1), $\rho = 0.7$	PD-feasible, clinically realistic
`c.bm` (Arch B)	$\{0, 0.25, 0.45\}$	Max PD-feasible ≈ 0.45 (OL+BDC)
`c.bm` (Arch A)	up to 0.95	PD constraint absent under Arch A
Carryover $t_{1/2}$	$\{0, 0.5, 1.0\}$ weeks	Clinically realistic
`lambda_cor`	$\ln 2/t_{1/2}$ (auto)	Pharmacokinetic derivation
Cross-factor (same t)	`c.cf1t` $= 0.2$	Unchanged from publication
Cross-factor (diff t)	`c.cfct` $\cdot \rho^{\lvert t_i - t_j \rvert}$	AR(1) decay applied
Analysis model	`nlme::lme` + `corCAR1`	Required for AR(1) Type I control
PD corrections	0/162 (0%)	All matrices valid

Parameter	Value	Issue
Autocorrelation	CS, rho = 0.8	PD failures at c.bm > 0.25
c.bm	{0, 0.3, 0.6}	0.6 exceeds PD-feasible range
Carryover t_half	{0, 0.1, 0.2} weeks	Too short for genuine effects
BM-BR correlation	Step function	Power drop artifact
Analysis model	lmer (no corCAR1)	Correct under CS
PD corrections	40/162 (24.7%)	Silent, uncontrolled